

Random maths

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Version	Date	Comments
0.1	October 7, 2026	First draft

Abstract

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1 Definitions

Use what you need in the symbols and functions defined in [Table 1](#), [Table 2](#), and [Table 3](#). Stay consistent with the lab’s notation and your own symbols.

Table 1: General symbol definitions.

Symbol	Explanation
a	A scalar
$\mathbf{v}$	A vector of D dimensions, $\mathbf{v} = [v_1, v_2, \dots, v_d]^T$
$\mathcal{S}$	A set of vectors with N elements in it, $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_1, \dots, \mathbf{v}_n\}$
$\mathbf{S}$	Matrix representation of a set $\mathcal{S}$ of size $D \times N$, $\mathbf{S} = [\mathbf{v}_1, \mathbf{v}_1, \dots, \mathbf{v}_n]$
$\mathbf{a} \cdot \mathbf{b} = c$	Dot product of two vectors, also known as inner product
$\mathbf{a}^T \mathbf{b} = c$	Inner product of two vectors, also known as dot product
$f(\cdot)$	A function, short version of $\text{function}(\cdot)$, same font as $\log$, $\exp$, $\arg \min$
$\ \mathbf{v}\ _2$	Euclidian distance or ℓ^2 -distance, $\ \mathbf{v}\ _2 = \sqrt{\mathbf{v}^T \mathbf{v}}$

Table 2: Symbol definitions for point clouds and registration.

<i>Symbol</i>	<i>Explanation</i>
d	Indices used for the dimension of both the point clouds with $d = \{1, 2, \dots, D\}$
i	Indices used for a point in the reading (moving) point cloud with $i = \{1, 2, \dots, I\}$
$\mathbf{p}_i$	A point in the reading point cloud, $\mathbf{p} = [x, y, z]^T$ for a 3D point
$\mathcal{P}$	A set of discrete sample points representing the reading (moving) point cloud, $\mathcal{P} = \{\mathbf{p}_1, \mathbf{p}_1, \dots, \mathbf{p}_i\}$
$\mathbf{P}$	Matrix version of $\mathcal{P}$ with a size of $D \times I$
$j, q_j, \mathcal{Q}, \mathbf{Q}$	Same definitions but for the reference (static) point cloud
k	Indices used when points are matched (points in $\mathbf{p}$ matched to $\mathbf{q}$) with $k = \{1, 2, \dots, K\}$
$\mathbf{e}_k$	Matched error between two points, $\mathbf{e}_k = \mathbf{p}'_i - \mathbf{q}_j$
$\mathcal{E}, \mathbf{E}$	Set and its matrix version of size $D \times K$ containing all matched error
$\mathbf{x}$	States of the robot
$\mathbf{x}_k$	States of the robot at the same time as the point $\mathbf{p}_k$
$\mathbf{p}'_k$	Moved reading point cloud as $\mathbf{p}'_k = \mathbf{T}(\mathbf{x}, \mathbf{p})$
$\mathbf{R}$	Rotation matrix of size $D \times D$
$\mathbf{T}$	Rigid transformation matrix of size $(D + 1) \times (D + 1)$
$\mathbf{n}_{q_k}$	Normal vector representing a plane at $\mathbf{p}_k$
$\mathbf{W}_{q_k}$	Covariance matrix of size $D \times D$ expressed in the reference frame of $\mathbf{q}_k$

Table 3: Function Definitions.

<i>Function</i>	<i>Explanation</i>
$\mathbf{T}(\mathbf{x}, \mathbf{p})$	Function transforming moving the reading point cloud
$\mathbf{T}(\mathbf{x}, \mathbf{p}_k)$	Rigid transformation with all points in the reading using the same states
$\mathbf{T}(\mathbf{x}_k, \mathbf{p}_k)$	Flexible transformation with each point moved by their own states
$\mathbf{R}(\mathbf{x})$	Function building a rotation matrix from the states
$\mathbf{J}(\mathbf{x}) = a$	Objective function also known as error function, loss function, cost function
$\text{match}(\mathcal{P}, \mathcal{Q}) = \mathcal{E}$	Matching function generating a set of error vectors $\mathcal{E}$ with K elements in it

2 Maths

2.1 State relations

Figure 1 shows the states of the robot and how to move from one to another.

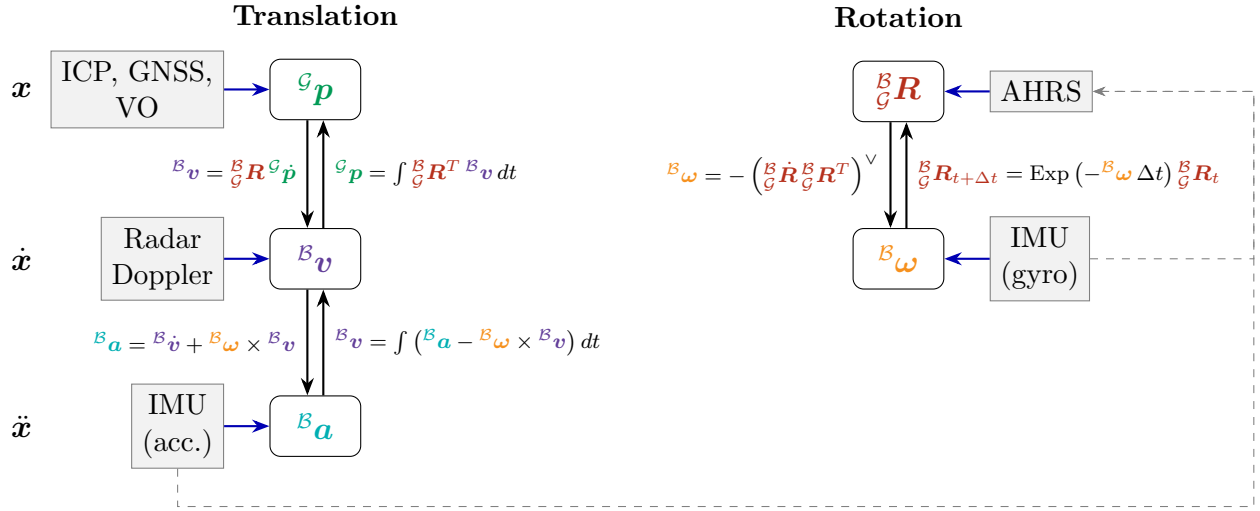


Figure 1: States of the robot. Down arrows derive and up arrows integrate between orders, blue arrows are direct measurements, and dashed arrows are dependencies. Left superscripts give the frame, global $\mathcal{G}$ or body $\mathcal{B}$, and ${}^{\mathcal{B}}_{\mathcal{G}}\mathbf{R}$ rotates vectors from $\mathcal{G}$ to $\mathcal{B}$. $\text{Exp}(\cdot)$ maps a rotation vector to $SO(3)$ and $(\cdot)^\vee$ maps a skew-symmetric matrix to its vector.

Curvature

$$\kappa = \frac{\Delta\theta}{\Delta s}$$

$$\kappa = \frac{1}{r}$$

Accelerometer model

$$\begin{aligned}\tilde{\mathbf{a}} &= \mathbf{a} + {}^{\mathcal{B}}_{\mathcal{G}}\mathbf{R}^{\mathcal{G}}\mathbf{g} + \boldsymbol{\eta}(t) + \mathbf{b}(t) \\ \mathbf{a} &= \mathbf{a}_c + \dots \\ \mathbf{a}_c &= \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \\ F &= ma\end{aligned}$$

$\mathbf{r}'$ is the position vector of the object relative to the rotating reference frame. [Wikipedia coriolis force](#)

Links using scalars

$$a_c = \frac{v^2}{r}$$

$$v = \omega r$$

$$v = \frac{\omega}{\kappa}$$